

A. Cagla ACUN

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Machine learning researcher developing explainable AI methods that make black-box models transparent without sacrificing accuracy, with applications spanning healthcare, cybersecurity, robotics, and education.

EDUCATION

MAY 2024 **Ph.D. Computer Science and Engineering - University of Louisville, KY, US**

Focus: Machine Learning - Advisor: [Dr. Olfa Nasraoui](#)

AUG 2018 **M.Sc. Computer Science and Engineering - University of Louisville, KY, US**

Focus: Machine Learning - Advisor: [Dr. Olfa Nasraoui](#)

JUN 2013 **B.Sc. Computer Engineering - Hacettepe University, Ankara, Turkey**

Honor Student

WORK EXPERIENCE

PRESENT **Independent Researcher**

MAY 2024

Conducting independent and collaborative research in explainable AI, Android malware detection, and machine learning for content classification.

- **XAI4Malware: A Comparative Evaluation of Explainable Artificial Intelligence Techniques for Android Malware Detection:** Researcher on a TUBITAK 1001 Scientific and Technological Research Projects Support Program proposal (PI: Prof. Dr. Sevil Sen, Hacettepe University) proposing a comprehensive and systematic evaluation of pre-hoc, co-hoc, LIME, and SHAP explainability paradigms for Android malware detection, including the construction of the first fine-grained malicious-code localization dataset. Proposal submitted in March 2026.
- **In-Training Explainability for Android Malware Detection [15]:** Adapted and implemented the Pre-hoc and Co-hoc in-training explainability frameworks for Android malware detection for the first time, benchmarking them against LIME and SHAP on the KronoDroid Real Dataset across fidelity, faithfulness, stability, sparsity, and computational efficiency. In-training methods produced more stable and reproducible explanations while maintaining 99.1% AUC, with Co-hoc running up to 77× faster than SHAP. Submitted to IEEE Transactions on Information Forensics and Security (under review).
- **Beyond Simple Blocklists: Measuring the Performance Gap Between Keyword and ML-Based Brand Safety Classification on Premium News Content [13]:** Quantified the performance gap between keyword-based and ML-based brand safety classification on premium news content by analyzing 83,258 articles from 12 publishers, introducing a two-construct framework that distinguishes topic avoidance from toxicity avoidance. GARM-aligned keyword blocklists flagged 44.1% of premium content versus under 0.2% for ML toxicity classifiers, demonstrating substantial overblocking of quality journalism.

NOV 2025 **Postdoctoral Fellow at Logistics and Distribution Institute (LoDI), University of Louisville**

DEC 2024

Developing AI frameworks and explainable machine learning methodologies to enhance transparency in clinical decision support systems and black-box algorithms.

- Led development of a novel decision support system integrating multi-criteria decision analysis with large language models to improve preoperative evaluation and optimization in orthopedic surgery (Proposal Submitted to United States National Institute of Health, collaboration with Dr. Lihui Bai and Dr. Jiapeng Huang).
- **Predicting Missing Criminal Links: Analyzing Organ Trafficking Networks [14]:** Developed a supervised link prediction pipeline to infer missing connections in organ trafficking networks, constructing a directed, role-aware country-level network from 2000–2022 incident data and combining local network structure signals with country-level indicators under a temporally ordered evaluation. Tree-based ensemble models were most effective, with local structural features dominating the predictions. Submitted to Expert Systems with Applications (under review).

Nov 2025 **Assistant Professor at Hitit University**

Nov 2024

Faculty member in the Department of Computer Engineering, teaching undergraduate courses and conducting research in machine learning and explainable AI.

MAY 2024 **Research Assistant at Web Mining and Knowledge Discovery Lab**

AUG 2015

I worked towards enabling explainability in machine learning to make black-box machine learning algorithms more transparent and trustworthy.

- **Evaluation of Refined Conversational Recommendation Based on Reprompting ChatGPT with Feedback** [8]: Conducted comprehensive research on evaluating Large Language Models (LLMs) effectiveness as a conversational recommendation system. Developed a novel evaluation pipeline to assess recommendation quality using user feedback and reprompting strategies. Demonstrated that iterative feedback improves recommendation relevancy and that popularity bias can be effectively managed using prompt engineering techniques. We built an evaluation pipeline around GPT-3.5-Turbo API that allows us to provide iterative feedback to simulate how a user would interact with it. Then, we evaluate popularity bias and compare ChatGPT's recommendations to baseline models. We found that re-prompting ChatGPT with iterative feedback had a significant impact on precision, and that popularity bias could be mitigated through prompt engineering.
- **Enhancing Robotic Grasp Failure Prediction Using A Pre-hoc Explainability Framework** [6], [9]: Led research on developing a novel pre-hoc explainability framework for robotic grasping failure prediction using Factorization Machines. Achieved 83% prediction accuracy while maintaining high explanation fidelity, demonstrating that pre-hoc methods can outperform traditional post-hoc approaches in both accuracy and transparency. The framework successfully identified key joint effort patterns that influence grasp failures in robotic manipulation.
- **GAI-Enhanced Assignment Framework: A Case Study on Generative AI Powered History Education** [4]: Developed and implemented a 'Reference-Check Protocol' for integrating Generative AI (ChatGPT) into undergraduate 'History of Science' curriculum, enhancing student engagement and learning outcomes through role-play exercises while addressing accuracy and authenticity challenges.
- **[Winner of the Best Student Paper Award] In-Training Explainability Frameworks: A Method to Make Black-Box Machine Learning Models More Explainable**[5], [11]: We introduced pre-hoc and co-hoc explainability approaches to enhance the interpretability of black-box models while maintaining prediction accuracy. Explanations were used as a regularizer during training, demonstrating advantages in terms of fidelity on three real-life datasets without compromising accuracy, in contrast to post-hoc methods.
- **A Data Science Approach to Flagging Non-Retention in Engineering Enrollment Data** [3]: Developed a versatile and scalable data science pipeline for retention analysis using enrollment data, with "three consecutive non-enrolled engineering semesters" as the optimal short-term metric, offering advantages over traditional methods reliant on grades or demographics.
- **A Data Science Pipeline for Educational Data: a Case Study Using Learning Catalytics in the Active Learning Classroom** [1][2]: Employed data mining and statistical techniques to uncover trends and peculiarities in educational data through a four-stage data science pipeline, revealing insights into learner behavior and course unit characteristics.

APR 2024 **Research Assistant at Louisville Automation and Robotics Research Institute (LARRI)**

AUG 2021

Developing machine learning algorithms and simulation as a part of NSF-funded project, Enhancing Future Work of Nursing Professionals through Collaborative Human-Robot Interfaces. Using PyTorch for machine learning development, Python, ROS, and Gazebo for robot and hospital simulations.

- **Task Allocation for Nursing Robots Using Factorization Machines in Explainable Machine Learning Framework** [10]: Developed a machine learning-based recommender system to optimize task allocation between robots and nurses in a hospital wing using a two-phased approach: simulating a hospital environment to collect data and building an explainable Factorization Machines framework, achieving 97% accuracy in predicting efficient task allocation while identifying task difficulty as the most important feature.
- **Robot Failure Mode Prediction with Deep Learning Sequence Models** [7]: Developed novel Recurrent Neural Network-based techniques for early and accurate prediction of object grasp failures, outperforming state-of-the-art models with over 80% accuracy and AUC while providing interpretability for informed decision-making, using rich sequential input data and limited grasper motion observations.
- **Comparative Analysis of Post-hoc Explainable Methods for Robotic Grasp Failure Prediction**:[12] Conducted comparative analysis of Tree-SHAP, LIME, and TreeInterpreter for robotic grasp failure prediction, revealing that TreeInterpreter offers superior computational efficiency while maintaining strong consistency with Tree-SHAP in identifying critical failure factors.

AUG 2017 **Graduate Student Assistant - REACH University of Louisville**

JUNE 2020

- Managed the Computer Resource Center and Academic Leadership Center, hiring, training, and evaluating tutoring staff, resolving computer hardware/software problems, scheduling the tutors working hours, liaising with designated course faculty, conducting test review sessions, presenting workshops that engage students, providing tutoring to students on an as needed basis, ensuring data is logged correctly into the usage tracking system, and promoting REACH services.

- Tutored undergrad computer science students both in-person and in group settings. Courses: CSE 130 Introduction to Programming Languages(C/C++), CSE 220 Object Oriented Program Design with Java, CSE 302 Data Structures.

JUNE 2013 Software Developer - INNOVA IT Solutions

AUG 2015 Collaborated in an Agile/Scrum environment to develop new features and resolve bugs throughout the software development life cycle, utilizing Java, Spring, Hibernate, PL/SQL, and SQL scripts to design user-friendly interfaces and deliver high-quality work in monthly iterations.

SEP 2012 Undergrad Student Researcher - Hacettepe University

JUNE 2013 Principal Investigator on a TUBITAK 2209-A University Students Research Projects grant (Advisor: Dr. Sevil Sen). Implemented a social network analysis method on scientific literature data, developing a social network analysis module for kaynakca.info, a reference management website, and performing a bibliometric analysis on the Bibliography of Turkey. The platform also serves as a networking website for scientists and researchers to share, research, and collaborate.

AUG 2012 Software Engineering Intern - Research Institute for Software Development (TUBITAK)

JUNE 2012 Developed a mobile application that works synchronously with the web-based application to take and save photos with their GPS coordinate information into the system. Android and PrimeFaces framework are used in mobile application.

AUG 2011 Software Engineering Intern - National Academic Network and Information Center

JUNE 2011 Worked on reporting and analyzing national academic networks. Wrote SQL scripts, and fixed the errors on current system.

AWARDS

2023 Best Student Paper Award. At IEEE WI-IAT'23 Conference.

2021 CSE Arthur M. Riehl Award. Given by J. B. Speed School of Engineering, University of Louisville.

2019 Best Poster Presentation Award. Given by Graduate Student Regional Research Conference Committee.

2014 Study Abroad Scholarship. Given by the Ministry of National Education, Republic of Turkey.

2012 TUBITAK 2209-A Student Project Grant. Given by The Scientific and Technological Research Council of Turkey.

PUBLICATIONS

- [1] A. C. Acun, J. L. Hieb, and O. Nasraoui, "Using a data science pipeline for course data: A case study analyzing, heterogeneous student data in two flipped classes," in *2019 ASEE Annual Conference & Exposition*, Tampa, Florida: ASEE Conferences, Jun. 2019. DOI: [10.18260/1-2--33492](https://doi.org/10.18260/1-2--33492).
- [2] A. C. Acun, O. Nasraoui, and J. L. Hieb, "Wip: Finding the right questions: Using data science to close the loop with classroom response systems," in *2019 ASEE Annual Conference & Exposition*, Tampa, Florida: ASEE Conferences, Jun. 2019. DOI: [10.18260/1-2--33619](https://doi.org/10.18260/1-2--33619).
- [3] M. Boujelbene, K. Damak, A. C. Acun, *et al.*, "A data-science approach to flagging non-retention in engineering enrollment data," in *2020 ASEE Virtual Annual Conference Content Access*, Virtual On line: ASEE Conferences, Jun. 2020. [Online]. Available: <https://peer.asee.org/33996>.
- [4] C. Acun and R. Acun, "Gai-enhanced assignment framework: A case study on generative ai powered history education," in *NeurIPS Workshop on Generative AI for Education (GAIED)*, New Orleans, Louisiana: NeurIPS, Dec. 2023. [Online]. Available: https://gaied.org/neurips2023/files/48/48%5C_paper.pdf.
- [5] C. Acun and O. Nasraoui, "In-training explainability frameworks: A method to make black-box machine learning models more explainable," in *2023 IEEE/WIC International Conference on Web Intelligence and Intelligent Agent Technology (WI-IAT)*, 2023, pp. 230–237. DOI: [10.1109/WI-IAT59888.2023.00036](https://doi.org/10.1109/WI-IAT59888.2023.00036).
- [6] C. Acun, A. Ashary, D. O. Popa, and O. Nasraoui, "Enhancing robotic grasp failure prediction using a pre-hoc explainability framework," in *2024 IEEE 20th International Conference on Automation Science and Engineering (CASE)*, 2024, pp. 1993–1998. DOI: [10.1109/CASE59546.2024.10711481](https://doi.org/10.1109/CASE59546.2024.10711481).
- [7] K. Damak, M. Boujelbene, C. Acun, *et al.*, "Robot failure mode prediction with deep learning sequence models," *Neural Computing and Applications*, Dec. 2024. DOI: [10.1007/s00521-024-10856-1](https://doi.org/10.1007/s00521-024-10856-1).
- [8] K. D. Spurlock, C. Acun, E. Saka, and O. Nasraoui, "Evaluation of refined conversational recommendation based on reprompting chatgpt with feedback," in *ACM KDD workshop on Generative AI for Recommender Systems and Personalization (GenAIRecP'24)*, 2024. [Online]. Available: <https://genai-personalization.github.io/assets/papers/GenAIRecP2024/Spurlock.pdf>.

- [9] C. Acun, A. Ashary, D. O. Popa, and O. Nasraoui, "Optimizing local explainability in robotic grasp failure prediction," *Electronics*, vol. 14, no. 12, 2025, ISSN: 2079-9292. DOI: [10.3390/electronics14122363](https://doi.org/10.3390/electronics14122363). [Online]. Available: <https://www.mdpi.com/2079-9292/14/12/2363>.
- [10] C. Acun, A. Ashary, D. O. Popa, and O. Nasraoui, "Task allocation for nursing robots using explainable machine learning with factorization machines," in *2025 IEEE 21st International Conference on Automation Science and Engineering (CASE)*, 2025, pp. 1044–1049. DOI: [10.1109/CASE58245.2025.11164103](https://doi.org/10.1109/CASE58245.2025.11164103).
- [11] C. Acun and O. Nasraoui, "Pre hoc and co hoc explainability: Frameworks for integrating interpretability into machine learning training for enhanced transparency and performance," *Applied Sciences*, vol. 15, no. 13, 2025, ISSN: 2076-3417. DOI: [10.3390/app15137544](https://doi.org/10.3390/app15137544). [Online]. Available: <https://www.mdpi.com/2076-3417/15/13/7544>.
- [12] A. Alvanpour, C. Acun, K. Spurlock, *et al.*, "Comparative analysis of post hoc explainable methods for robotic grasp failure prediction," *Electronics*, vol. 14, no. 9, 2025, ISSN: 2079-9292. DOI: [10.3390/electronics14091868](https://doi.org/10.3390/electronics14091868). [Online]. Available: <https://www.mdpi.com/2079-9292/14/9/1868>.
- [13] C. Acun and I. Acun, "Beyond simple blocklists: Measuring the performance gap between keyword and ml-based brand safety classification on premium news content," in *International Conference on the AI Revolution: Research, Ethics, and Society (AIR-RES 2026)*, Las Vegas, NV, USA, Apr. 2026.
- [14] H. Balasuriya, C. Acun, M. Gentili, and N. Koizumi, "Predicting missing criminal links: Analyzing organ trafficking networks," Submitted to Expert Systems with Applications (under review), 2026.
- [15] O. Ceviz, C. Acun, and S. Sen, "In-training explainability for android malware detection," Submitted to IEEE Transactions on Information Forensics and Security (under review), 2026.